

## CATARMAN, Philippines

WMO: 985460

Lat: 12.5056N

Lon: 124.6285E

Elev: 2

StdP: 101.30

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | 22.5       | 23.2 | 20.9                          | 15.5 | 24.0 | 21.5 | 16.2 | 24.4 | 10.2                  | 24.7 | 8.5 | 25.2 | 1.2                   | 180  | 0.342 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 5             | 7.1                    | 33.7            | 27.7 | 33.1 | 27.7 | 32.6 | 27.6 | 28.6                | 32.0 | 28.3 | 31.7 | 28.1 | 31.5 | 2.9                  | 220  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 27.6                            | 23.6 | 30.9 | 27.3 | 23.2 | 30.7 | 27.1 | 22.9 | 30.5 | 92.6          | 32.3 | 91.2 | 32.0 | 90.0 | 31.8 | 29.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| 7.7               | 6.4  | 5.5 | 21.4                       | 34.9 | 1.1                | 0.6 | 20.6                                               | 35.3 | 20.0       | 35.6 | 19.4       | 35.9 | 18.6       | 36.4 |    |    |
|                   |      |     | 20.4                       | 29.1 | 1.3                | 0.5 | 19.4                                               | 29.5 | 18.6       | 29.8 | 17.9       | 30.0 | 16.9       | 30.4 |    |    |

## Monthly Climatic Design Conditions

|      |                                                                   | Annual      | Jan    | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|-------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                   | (d)         | (e)    | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 27.5   | 26.2  | 26.4  | 26.8  | 27.7  | 28.3  | 28.3  | 28.0  | 28.4  | 28.1  | 27.7  | 27.3  | 26.7 | (6)  |
| (7)  |                                                                   | DBStd       | 1.22   | 0.87  | 0.94  | 0.88  | 0.84  | 0.86  | 1.07  | 1.09  | 1.08  | 1.04  | 0.94  | 0.90  | 0.94 | (7)  |
| (8)  |                                                                   | HDD10.0     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | (8)  |
| (9)  |                                                                   | HDD18.3     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | (9)  |
| (10) |                                                                   | CDD10.0     | 6390   | 502   | 459   | 522   | 531   | 569   | 549   | 559   | 571   | 544   | 548   | 518   | 517  | (10) |
| (11) |                                                                   | CDD18.3     | 3348   | 244   | 226   | 263   | 281   | 311   | 299   | 301   | 312   | 294   | 290   | 268   | 259  | (11) |
| (12) |                                                                   | CDH23.3     | 35202  | 2043  | 2016  | 2574  | 3120  | 3693  | 3420  | 3327  | 3551  | 3248  | 3099  | 2723  | 2390 | (12) |
| (13) |                                                                   | CDH26.7     | 11371  | 310   | 420   | 704   | 1089  | 1491  | 1337  | 1224  | 1375  | 1190  | 1003  | 758   | 468  | (13) |
| (14) | Wind                                                              | WSAvg       | 2.2    | 3.4   | 2.9   | 2.6   | 2.1   | 1.7   | 1.6   | 1.7   | 1.9   | 1.7   | 1.9   | 2.3   | 3.1  | (14) |
| (15) | Precipitation                                                     | PrecAvg     | 3569   | 468   | 296   | 286   | 162   | 160   | 221   | 227   | 172   | 224   | 318   | 444   | 590  | (15) |
| (16) |                                                                   | PrecMax     | 6754   | 1419  | 1191  | 1304  | 607   | 784   | 457   | 448   | 380   | 493   | 578   | 1001  | 1015 | (16) |
| (17) |                                                                   | PrecMin     | 1981   | 117   | 37    | 38    | 15    | 26    | 86    | 53    | 44    | 51    | 148   | 101   | 116  | (17) |
| (18) |                                                                   | PrecStd     | 1102   | 264   | 285   | 274   | 146   | 150   | 103   | 107   | 80    | 105   | 124   | 161   | 240  | (18) |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 30.2  | 31.3  | 32.0  | 32.6  | 34.1  | 34.5  | 33.9  | 34.5  | 33.9  | 33.1  | 31.9  | 31.0 | (19) |
| (20) |                                                                   |             | MCWB   | 26.9  | 27.0  | 28.0  | 27.9  | 28.0  | 27.7  | 27.8  | 27.5  | 27.7  | 27.6  | 27.4  | 27.5 | (20) |
| (21) |                                                                   | 2%          | DB     | 29.6  | 30.3  | 31.1  | 32.0  | 33.1  | 33.5  | 33.0  | 33.3  | 33.0  | 32.1  | 31.2  | 30.2 | (21) |
| (22) |                                                                   |             | MCWB   | 26.5  | 26.7  | 27.2  | 27.6  | 27.9  | 27.8  | 27.6  | 27.5  | 27.5  | 27.5  | 27.4  | 27.1 | (22) |
| (23) |                                                                   | 5%          | DB     | 29.0  | 29.7  | 30.5  | 31.5  | 32.5  | 32.7  | 32.2  | 32.6  | 32.2  | 31.4  | 30.6  | 29.7 | (23) |
| (24) |                                                                   |             | MCWB   | 26.2  | 26.3  | 26.8  | 27.4  | 27.8  | 27.7  | 27.4  | 27.4  | 27.4  | 27.3  | 27.2  | 26.8 | (24) |
| (25) |                                                                   | 10%         | DB     | 28.4  | 29.0  | 29.9  | 31.0  | 31.8  | 31.8  | 31.5  | 31.8  | 31.5  | 30.7  | 30.0  | 29.0 | (25) |
| (26) |                                                                   |             | MCWB   | 25.9  | 25.9  | 26.4  | 27.2  | 27.6  | 27.6  | 27.2  | 27.2  | 27.2  | 27.1  | 27.0  | 26.6 | (26) |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 27.7  | 27.7  | 28.6  | 28.6  | 28.7  | 28.7  | 28.6  | 28.6  | 28.6  | 28.5  | 28.4  | 28.2 | (27) |
| (28) |                                                                   |             | MCD B  | 29.5  | 30.1  | 31.4  | 31.7  | 32.3  | 32.5  | 32.4  | 32.5  | 32.4  | 31.6  | 30.6  | 30.0 | (28) |
| (29) |                                                                   | 2%          | WB     | 27.1  | 27.2  | 27.7  | 28.1  | 28.5  | 28.3  | 28.1  | 28.2  | 28.1  | 28.0  | 27.8  | 27.6 | (29) |
| (30) |                                                                   |             | MCD B  | 28.9  | 29.6  | 30.4  | 31.2  | 32.1  | 32.1  | 31.9  | 32.0  | 31.8  | 31.1  | 30.3  | 29.5 | (30) |
| (31) |                                                                   | 5%          | WB     | 26.6  | 26.8  | 27.2  | 27.7  | 28.1  | 28.0  | 27.7  | 27.8  | 27.7  | 27.6  | 27.5  | 27.2 | (31) |
| (32) |                                                                   |             | MCD B  | 28.4  | 29.0  | 29.8  | 30.8  | 31.7  | 31.9  | 31.3  | 31.5  | 31.3  | 30.6  | 29.9  | 29.1 | (32) |
| (33) |                                                                   | 10%         | WB     | 26.2  | 26.3  | 26.7  | 27.4  | 27.8  | 27.7  | 27.4  | 27.4  | 27.4  | 27.2  | 27.1  | 26.7 | (33) |
| (34) |                                                                   |             | MCD B  | 27.8  | 28.4  | 29.3  | 30.4  | 31.3  | 31.3  | 30.9  | 31.1  | 30.8  | 30.2  | 29.5  | 28.5 | (34) |
| (35) | Mean Daily Temperature Range                                      | MDBR        | 4.0    | 5.0   | 5.9   | 6.5   | 7.1   | 6.9   | 6.5   | 6.6   | 6.5   | 5.9   | 5.3   | 4.2   | (35) |      |
| (36) |                                                                   | 5% DB       | MCD BR | 5.3   | 6.3   | 7.2   | 7.3   | 8.1   | 8.1   | 7.7   | 7.8   | 7.7   | 7.1   | 6.5   | 5.4  | (36) |
| (37) |                                                                   |             | MCWBR  | 3.1   | 3.7   | 4.3   | 4.1   | 4.2   | 3.9   | 3.7   | 3.5   | 3.6   | 3.6   | 3.6   | 3.3  | (37) |
| (38) |                                                                   | 5% WB       | MCD BR | 4.8   | 5.8   | 6.6   | 7.0   | 7.5   | 7.6   | 7.2   | 7.1   | 6.6   | 6.1   | 5.2   | (38) |      |
| (39) |                                                                   |             | MCWBR  | 3.3   | 3.7   | 4.3   | 4.2   | 4.2   | 4.0   | 3.9   | 3.8   | 3.8   | 3.7   | 3.8   | 3.5  | (39) |
| (40) | Clear-Sky Solar Irradiance                                        | taub        | 0.466  | 0.442 | 0.440 | 0.429 | 0.429 | 0.427 | 0.432 | 0.458 | 0.448 | 0.454 | 0.450 | 0.469 | (40) |      |
| (41) |                                                                   | taud        | 2.273  | 2.355 | 2.375 | 2.410 | 2.438 | 2.455 | 2.440 | 2.351 | 2.379 | 2.354 | 2.357 | 2.268 | (41) |      |
| (42) |                                                                   | Ebn at Noon | 836    | 874   | 881   | 881   | 867   | 859   | 857   | 845   | 861   | 852   | 845   | 821   | (42) |      |
| (43) |                                                                   | Edh at Noon | 133    | 127   | 127   | 122   | 116   | 113   | 115   | 127   | 124   | 125   | 122   | 131   | (43) |      |
| (44) | All-Sky Solar Radiation                                           | RadAvg      | 3.83   | 4.79  | 5.41  | 6.08  | 5.84  | 5.28  | 4.83  | 5.12  | 4.93  | 4.56  | 4.26  | 3.70  | (44) |      |
| (45) |                                                                   | RadStd      | 0.53   | 0.62  | 0.64  | 0.44  | 0.47  | 0.30  | 0.41  | 0.29  | 0.45  | 0.31  | 0.39  | 0.38  | (45) |      |

## Historical Trends

|      |                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (a)  | (b)                    | (c)   | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) | Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (7 neighbors) | +0.18 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page